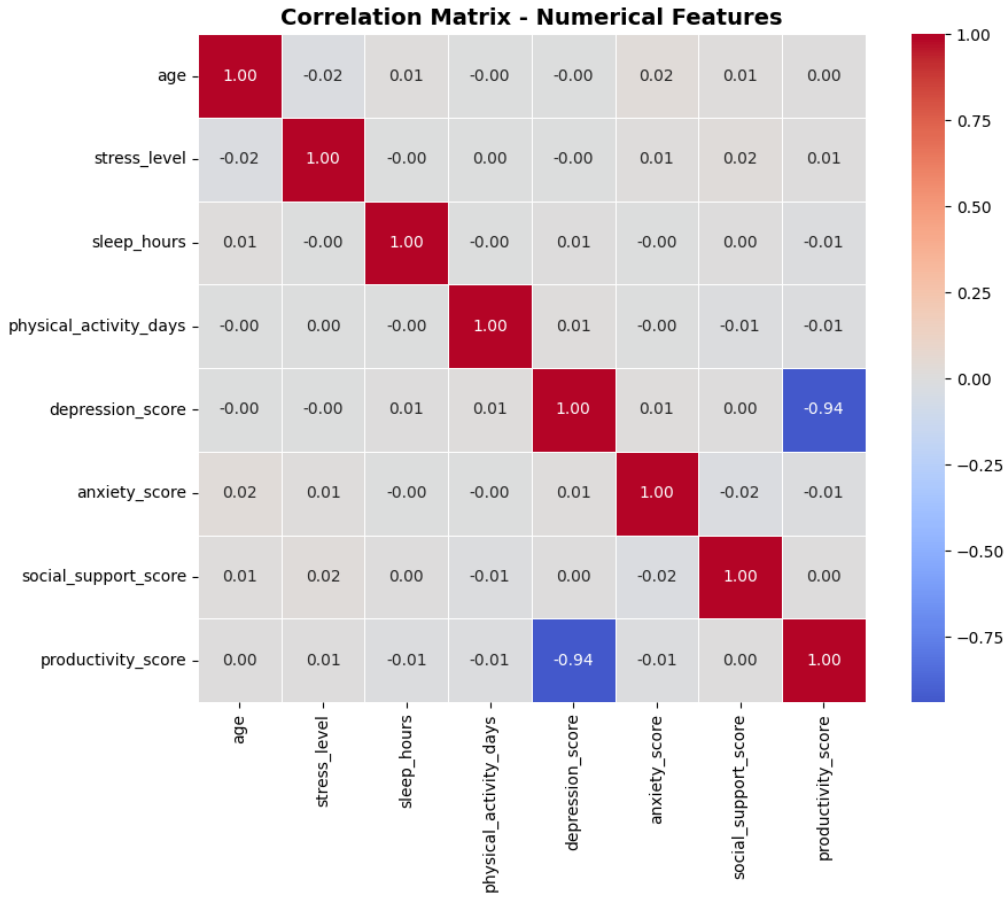
CORRELATION ANALYSIS - Mental Health Dataset

1. NUMERICAL FEATURES CORRELATIONS

Numerical Features Correlation Matrix:

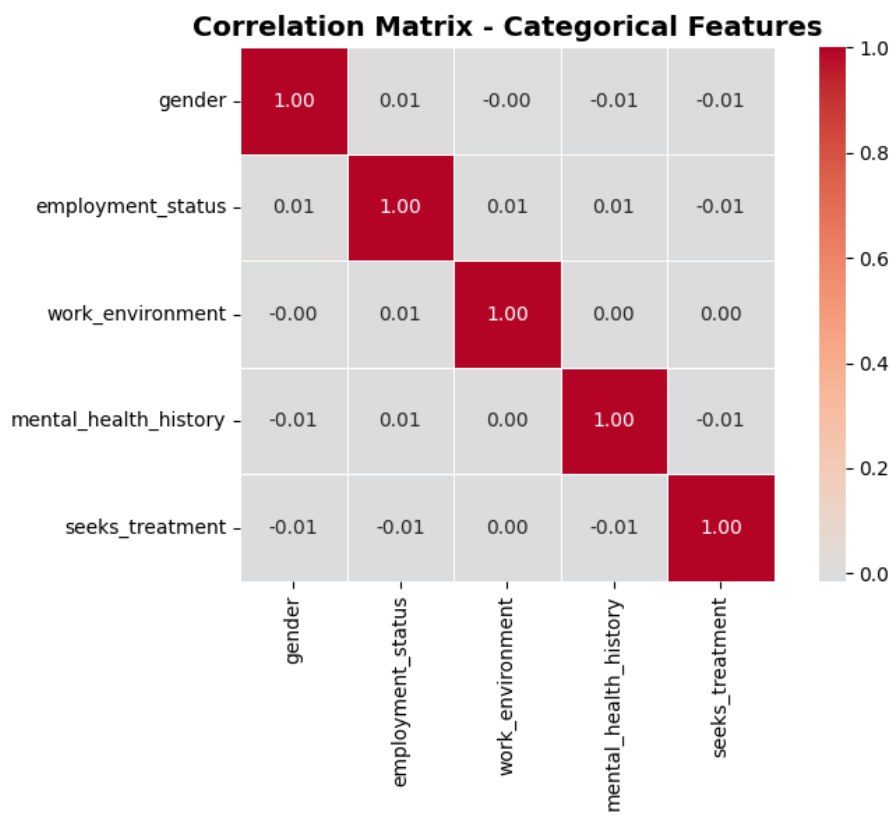
Feature ↓ / Feature →	Age	Stress Level	Sleep Hours	Physical Activity Days	Depression Score	Anxiety Score	Social Support Score	Productivity Score
Age	1.000	-0.016	0.008	-0.004	-0.001	0.025	0.011	0.002
Stress Level	-0.016	1.000	-0.003	0.000	-0.002	0.012	0.020	0.007
Sleep Hours	0.008	-0.003	1.000	-0.001	0.005	-0.004	0.003	-0.009
Physical Activity Days	-0.004	0.000	-0.001	1.000	0.009	-0.002	-0.010	-0.010
Depression Score	-0.001	-0.002	0.005	0.009	1.000	0.011	0.003	-0.939
Anxiety Score	0.025	0.012	-0.004	-0.002	0.011	1.000	-0.018	-0.009
Social Support Score	0.011	0.020	0.003	-0.010	0.003	-0.018	1.000	0.001
Productivity Score	0.002	0.007	-0.009	-0.010	-0.939	-0.009	0.001	1.000


Correlation Matrix - Numerical Features
age
stress_level
sleep_hours
physical_activity_days
depression_score
anxiety_score
social_support_score
productivity_score
1.00 -0.02 0.01 -0.00 -0.00 0.02 0.01 0.00
-0.02 1.00 -0.00 0.00 -0.00 0.01 0.02 0.01
0.01 -0.00 1.00 -0.00 0.01 -0.00 0.00 -0.01
-0.00 0.00 -0.00 1.00 0.01 -0.00 -0.01 -0.01
-0.00 -0.00 0.01 0.01 1.00 0.01 0.00 -0.94
0.02 0.01 -0.00 -0.00 0.01 1.00 -0.02 -0.01
0.01 0.02 0.00 -0.01 0.00 -0.02 1.00 0.00
0.00 0.01 -0.01 -0.01 -0.94 -0.01 0.00 1.00
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75

2. CATEGORICAL FEATURES CORRELATIONS (Label Encoded)

Categorical Features Correlation Matrix:

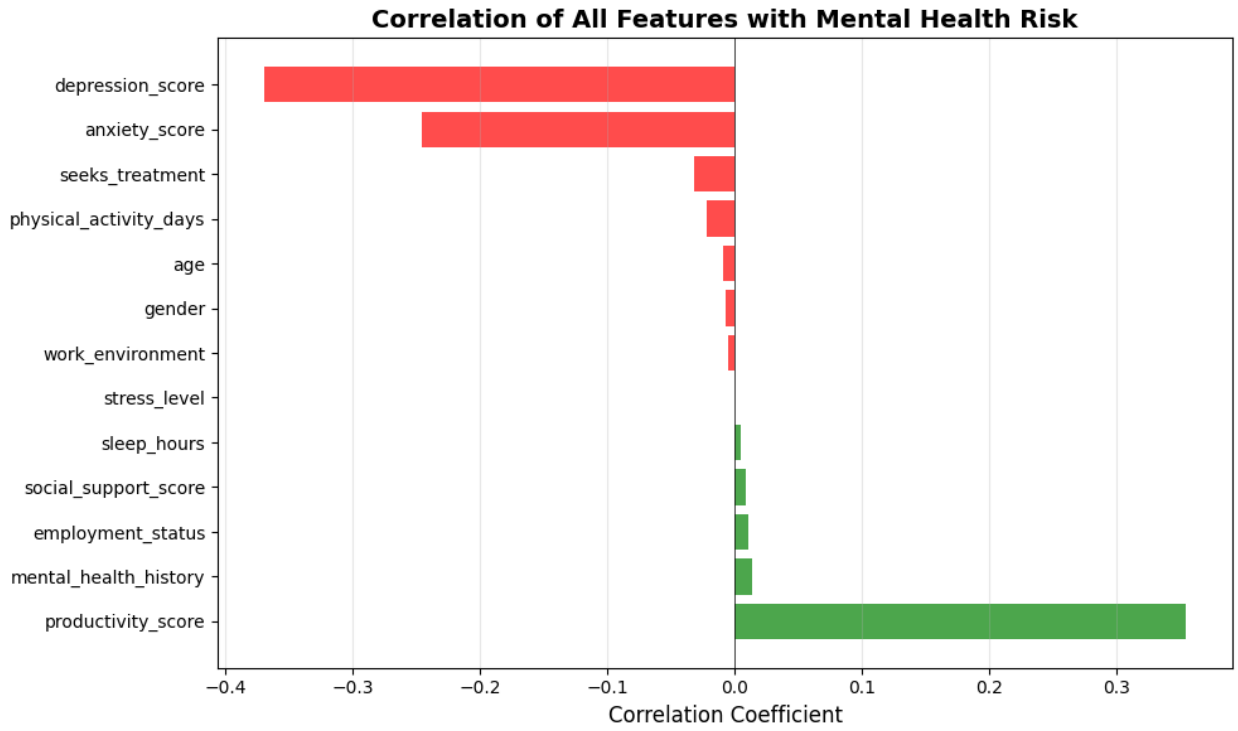
Feature ↓ / Feature →	Gender	Employment Status	Work Environment	Mental Health History	Seeks Treatment
Gender	1.000	0.011	-0.005	-0.013	-0.013
Employment Status	0.011	1.000	0.009	0.008	-0.010
Work Environment	-0.005	0.009	1.000	0.002	0.004
Mental Health History	-0.013	0.008	0.002	1.000	-0.015
Seeks Treatment	-0.013	-0.010	0.004	-0.015	1.000



3. CORRELATION OF ALL X FEATURES WITH TARGET (mental_health_risk)

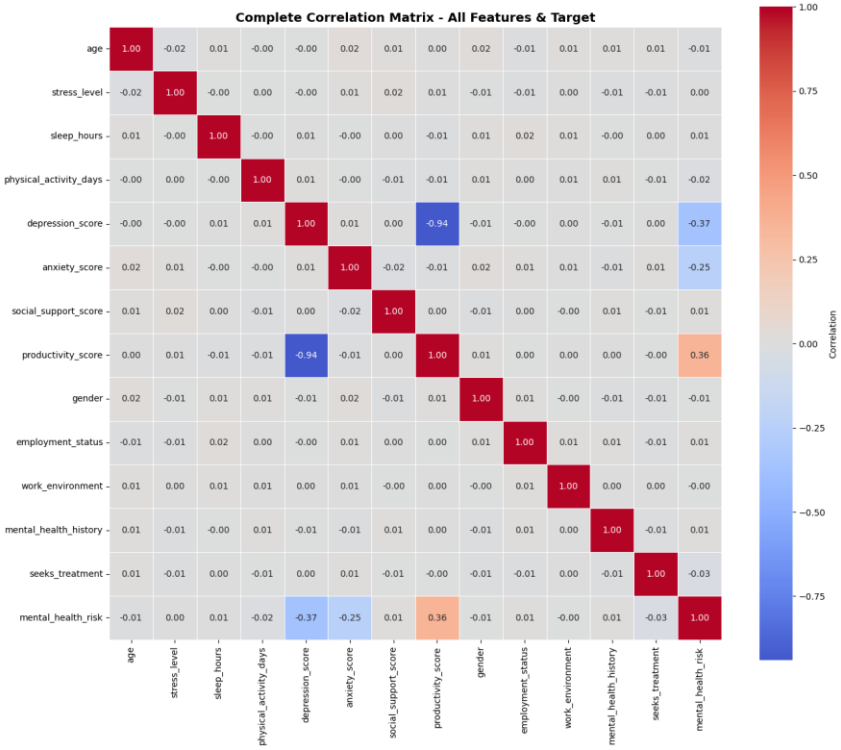
Correlation with Target (mental_health_risk):

Feature	Correlation Value
Productivity Score	0.355
Mental Health History	0.014
Employment Status	0.011
Social Support Score	0.010
Sleep Hours	0.005
Stress Level	0.000
Work Environment	-0.004
Gender	-0.006
Age	-0.009
Physical Activity Days	-0.022
Seeks Treatment	-0.032
Anxiety Score	-0.246
Depression Score	-0.370


Correlation of All Features with Mental Health Risk
depression_score
anxiety_score
seeks_treatment
physical_activity_days
age
gender
work_environment
stress_level
sleep_hours
social_support_score
employment_status
mental_health_history
productivity_score
−0.4
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
Correlation Coefficient

4. COMBINED CORRELATION MATRIX (All Features + Target)

Feature ↓ / Feature →	Age	Stress	Sleep	Phys. Act.	Depress.	Anxiety	Social Sup.	Productiv	Gender	Employ.	Work Env.	MH Hist.	Treat.	MH Risk
Age	1.00	-0.02	0.01	-0.00	-0.00	0.03	0.01	0.00	0.02	-0.01	0.01	0.01	0.01	-0.01
Stress Level	-0.02	1.00	-0.00	0.00	-0.00	0.01	0.02	0.01	-0.01	-0.01	0.00	-0.01	-0.01	0.00
Sleep Hours	0.01	-0.00	1.00	-0.00	0.01	-0.00	0.00	-0.01	0.01	0.02	0.01	-0.00	0.00	0.01
Physical Activity Days	-0.00	0.00	-0.00	1.00	0.01	-0.00	-0.01	-0.01	0.01	0.00	0.01	0.01	-0.01	-0.02
Depression Score	-0.00	-0.00	0.01	0.01	1.00	0.01	0.00	-0.94	-0.01	-0.00	0.00	-0.01	0.00	-0.37
Anxiety Score	0.03	0.01	-0.00	-0.00	0.01	1.00	-0.02	-0.01	0.02	0.01	0.01	-0.01	0.01	-0.25
Social Support Score	0.01	0.02	0.00	-0.01	0.00	-0.02	1.00	0.00	-0.01	0.00	-0.00	0.01	-0.01	0.01
Productivity Score	0.00	0.01	-0.01	-0.01	-0.94	-0.01	0.00	1.00	0.01	0.00	0.00	0.00	-0.00	0.36
Gender	0.02	-0.01	0.01	0.01	-0.01	0.02	-0.01	0.01	1.00	0.01	-0.01	-0.01	-0.01	-0.01
Employment Status	-0.01	-0.01	0.02	0.00	-0.00	0.01	0.00	0.00	0.01	1.00	0.01	0.01	-0.01	0.01
Work Environment	0.01	0.00	0.01	0.01	0.00	0.01	-0.00	0.00	-0.01	0.01	1.00	0.00	0.00	-0.00
Mental Health History	0.01	-0.01	-0.00	0.01	-0.01	-0.01	0.01	0.00	-0.01	0.01	0.00	1.00	-0.02	0.01
Seeks Treatment	0.01	-0.01	0.00	-0.01	0.00	0.01	-0.01	-0.00	-0.01	-0.01	0.00	-0.02	1.00	-0.03
Mental Health Risk	-0.01	0.00	0.01	-0.02	-0.37	-0.25	0.01	0.36	-0.01	0.01	-0.00	0.01	-0.03	1.00


Complete Correlation Matrix - All Features & Target
age 1.00 -0.02 0.01 -0.00 -0.00 0.02 0.01 0.00 0.02 -0.01 0.01 0.01 0.01 -0.01
stress_level -0.02 1.00 -0.00 0.00 -0.00 0.01 0.02 0.01 -0.01 -0.01 0.00 -0.01 -0.01 0.00
sleep_hours 0.01 -0.00 1.00 -0.00 0.01 -0.00 0.00 -0.01 0.01 0.02 0.01 -0.00 0.00 0.01
physical_activity_days -0.00 0.00 -0.00 1.00 0.01 -0.00 -0.01 -0.01 0.01 0.00 0.01 0.01 -0.01 -0.02
depression_score -0.00 -0.00 0.01 0.01 1.00 0.01 0.00 -0.94 -0.01 -0.00 0.00 -0.01 0.00 -0.37
anxiety_score 0.02 0.01 -0.00 -0.00 0.01 1.00 -0.02 -0.01 0.02 0.01 0.01 -0.01 0.01 -0.25
social_support_score 0.01 0.02 0.00 -0.01 0.00 -0.02 1.00 0.00 -0.01 0.00 -0.00 0.01 -0.01 0.01
productivity_score 0.00 0.01 -0.01 -0.01 -0.94 -0.01 0.00 1.00 0.01 0.00 0.00 0.00 -0.00 0.36
gender 0.02 -0.01 0.01 0.01 -0.01 0.02 -0.01 0.01 1.00 0.01 -0.00 -0.01 -0.01 -0.01
employment_status -0.01 -0.01 0.02 0.00 -0.00 0.01 0.00 0.00 0.01 1.00 0.01 0.01 -0.01 0.01
work_environment 0.01 0.00 0.01 0.01 0.00 0.01 -0.00 0.00 -0.00 0.01 1.00 0.00 0.00 -0.00
mental_health_history 0.01 -0.01 -0.00 0.01 -0.01 -0.01 0.01 0.00 -0.01 0.01 0.00 1.00 -0.01 0.01
seeks_treatment 0.01 -0.01 0.00 -0.01 0.00 0.01 -0.01 -0.00 -0.01 -0.01 0.00 -0.01 1.00 -0.03
mental_health_risk -0.01 0.00 0.01 -0.02 -0.37 -0.25 0.01 0.36 -0.01 0.01 -0.00 0.01 -0.03 1.00
age
stress_level
sleep_hours
physical_activity_days
depression_score
anxiety_score
social_support_score
productivity_score
gender
employment_status
work_environment
mental_health_history
seeks_treatment
mental_health_risk
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
Correlation

5. KEY INSIGHTS

Top 5 Features Most Correlated with Mental Health Risk (positive):

Metric	**Productivity Score**	**Mental Health History**	**Employment Status**	**Social Support Score**	**Sleep Hours**
Correlation with Mental Health Risk	**0.36**	0.01	0.01	0.01	0.01

Top 5 Features Most Correlated with Mental Health Risk (negative):

Metric	**Age**	**Physical Activity Days**	**Seeks Treatment**	**Anxiety Score**	**Depression Score**
Correlation with Mental Health Risk	-0.01	-0.02	-0.03	**-0.25**	**-0.37**

Strongest Correlations Between Features (excluding target):

Feature 1	**Feature 2**	**Correlation**
Depression Score	Productivity Score	**-0.94**
Depression Score	Mental Health Risk	-0.37
Productivity Score	Mental Health Risk	0.36
Anxiety Score	Mental Health Risk	-0.25
Seeks Treatment	Mental Health Risk	-0.03
Age	Anxiety Score	0.03
Physical Activity Days	Mental Health Risk	-0.02
Stress Level	Social Support Score	0.02
Sleep Hours	Employment Status	0.02
Anxiety Score	Social Support Score	-0.02